Song 2022 Prostate Cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

14/15 samples clear the ≥100-cell floor and intermix per-neighbourhood (iLISI median 1.95 → 7.12 of 14 after Harmony θ=10); one continuous manifold, 0 detached islands.

Gate 2 — module geography → GO (non-branching)

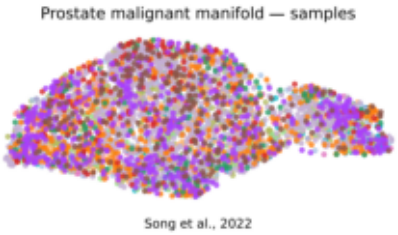
Dominant inverse gradient pSom↔Crest (r=−0.85); Gast↔pSom weak (r=−0.19). Tree tips all share one program → spurious forks; curve fold-free (self-approach 2.93). Single pSom→crest axis.

Gate 3 — hub mutual exclusivity → NA

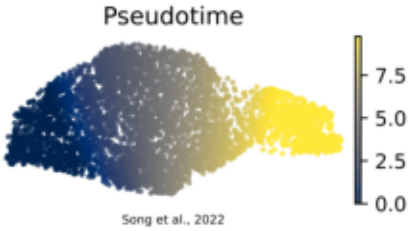
No bifurcation, so no convergence hub to test. Joint pSom&Crest-high enrichment 0.27 (≪1) confirms an inverse gradient, not a co-expression hub.

STEP FIGURES

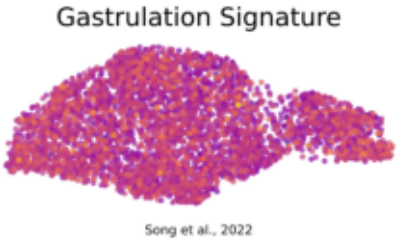
Manifold — samples



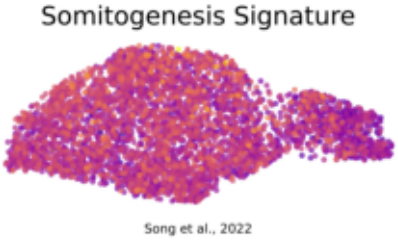
Pseudotime



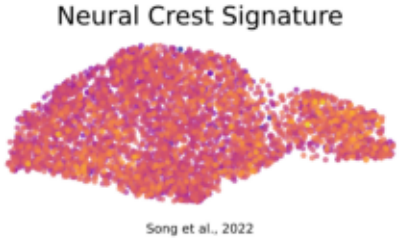
Gastrulation



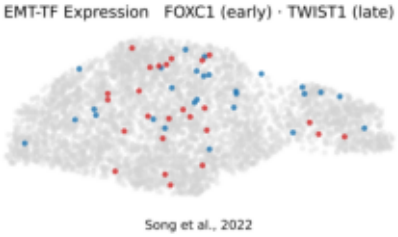
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Prostate Cancer (Song et al., 2022), Seq-Well S3 — 15 samples / 21,743 cells (master atlas); 4,658 malignant cells, of which 14 samples clear the ≥100-cell floor → 4,574 cells on the whole-manifold trajectory. Uniformly luminal AR-driven adenocarcinoma (KLK3 in 97% of cells). Two AR-output compartments: Luminal-AR-active (3,916) and Luminal-AR-low (658). Cycling 0.7%, orthogonal to all developmental programs.

Verdict.

Non-branching linear trajectory on a somitogenesis→neural-crest embryonic axis. Root at the Gast/pSom-high, Crest-low tip; pseudotime runs Gast/pSom-high → Crest-high (Crest decile mean pseudotime 0.95, terminal). Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. Classical mesenchymal EMT-TFs are largely below detection (max FOXC1 2.4%) — the honest low-EMT readout for a luminal epithelial adenocarcinoma; FOXC1 (early) / TWIST1 (late) shown.

Parameter choices

Min-cell threshold	≥100 floor (mult. gap degenerate at rank 0); 14/15 kept	**Integration**	Harmony on sample θ=10 (plateaus by θ=5); PCA 50→20
Overdispersed genes	199; neighbours k=30; UMAP spread=1.3, seed=1	**Islands / lobes**	UMAP radius graph 6× median-NN (0 removed); Leiden res=0.15
Trajectory	curve 20 nodes, epg λ=0.3 μ=100 (fold-free 2.93)	**Root**	Gast/pSom-high tip (Crest-low origin)
EMT-TF axis	FOXC1 (early) → TWIST1 (late)	**Geometry**	Linear — pSom/Crest (non-branching)